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Sim-to-Real Performance Analysis of a Custom Quadruped Robot Using Gait-Generator-Guided Reinforcement Learning

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Abstract—This paper analyzes the Sim-to-Real performance of a custom quadruped robot across foot motion, joint tracking, body stability, and actuator loading. A pure-RL controller produced a repeatable gait in simulation but failed before completing a walking cycle on hardware. The simulated mean swing peak was 7.54 cm, while the largest pre-protection front-foot displacement was only 2.57–3.53 cm; the front-right thigh current command and measured current reached 47.99 and 46.98 A within 0.675 s. We interpret this as a feasibility mismatch in which Pure RL without an explicit gait prior can find simulation-valid motions outside the smaller region executable by the physical robot. Rather than attempting to eliminate all hardware uncertainty, we first verified a robot-specific gait-generator reference by direct hardware execution and then used it to constrain PPO training. Two guided controllers were each evaluated in three hardware recordings at a 0.4 m/s level-ground forward command. Simulated mean swing peaks of 8.54–8.63 cm became 7.25–7.78 cm on hardware, corresponding to 84.9–90.0% amplitude retention and 0.980–0.990 phase correlation. Within this tested condition, gait guidance substantially improved practical transfer, while residual gaps remain in trajectory amplitude, joint execution, body dynamics, and actuator loading.

Index Terms—Quadruped robot, reinforcement learning, imitation learning, gait generator, sim-to-real transfer, foot clearance, joint tracking, body stability, actuator protection.

I. INTRODUCTION

Reinforcement learning has enabled quadruped robots to learn locomotion directly from interaction in simulation [1]–[6]. The main challenge in deployment is that the learned controller must operate on a physical mechanism whose friction, backlash, contact dynamics, sensing, structural compliance, and actuator limits are only approximated by the simulator [4], [10], [11]. For a custom-built robot, these discrepancies can appear simultaneously in foot trajectories, joint tracking, body motion, and actuator loading. Our study therefore focuses on the overall Sim-to-Real behavior of the locomotion controller rather than on a single performance index. Among the available measurements, foot clearance is the most direct indicator of successful swing-leg execution and was also the most urgent problem observed in our early hardware tests, so it is used as the principal trajectory metric throughout the paper.

Our initial development used pure RL. In simulation, the learned policies could produce repeatable forward gaits, but hardware tests repeatedly showed a much larger loss of swing-leg motion than expected. The feet remained close to the floor, toe drag prevented normal forward progression, the body shifted onto the front legs, and the front-thigh actuator could enter over-current protection. We first attempted to address the problem within the pure-RL formulation by adjusting foot-clearance targets, reward weights, orientation penalties, contact terms, and the standing pose. These changes improved individual simulations but did not reliably remove the hardware failure. The final pure-RL record analyzed in this paper therefore provides a useful example of a controller that appears acceptable in simulation but fails after transfer.

The failure suggested a mismatch between the broad set of motions that can satisfy the simulated task and the smaller subset that the physical robot can execute reliably. Assembly error, drivetrain loss, backlash, contact uncertainty, sensing error, and actuator limits cannot be eliminated completely, so eliminating enough physical and modeling uncertainty to make the broad simulation-admissible solution space reliably transferable was impractical for the current platform. We therefore constrained the learning search space instead. A robot-specific gait generator was adapted using the present link dimensions, joint conventions, and inverse kinematics (IK), and its generated motion was first executed directly on hardware without RL to verify feasibility. Joint- and foot-tracking references were then incorporated into PPO while retaining task rewards for velocity tracking, stabilization, contact behavior, smoothness, and effort. Unlike demonstration-driven imitation such as [7], the generator supplies a kinematic prior rather than a recorded expert motion; RL is retained for task optimization and adaptation.

II. CUSTOM QUADRUPED AND CONTROL STACK

A. Robot and Actuation

The robot has 12 actuated joints: hip, thigh, and calf on each leg. CubeMars RO80 motors are used with 10:1 gearboxes. The kinematic model uses 250 mm thigh and calf links and an 84 mm lateral hip link. Most housings, shafts, and transmission parts were made in the laboratory, so backlash, eccentricity, and friction were not identical from joint to joint. The calf also has an extra 2:1 reduction stage, which changes the effective stiffness seen at the output. For feedback, each joint uses encoder measurements for joint position and velocity, while ODrive current feedback is recorded to monitor

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actuator loading. An HWT906P IMU provides body orientation, angular velocity, and linear acceleration for state estimation and body-motion analysis.

Before deployment, the motors were checked in velocity control and ranked by current demand. Motors with lower mechanical resistance were assigned to the more heavily loaded joints. A dual-encoder magnet arrangement was removed after repeated rubbing and over-current events. We also reduced the calf gain because the same gain used on all joints caused visible vibration. ODrive firmware and velocity-limit changes removed recurring hip protection events, and rear-hip zero offsets were corrected before the final walking tests.



Fig. 1. Custom quadruped robot used for the Sim-to-Real experiments.

TABLE I. CONTROL-ORIENTED HARDWARE CALIBRATION

Item	Observed issue	Final control response
Calf drive	Extra 2:1 stage; vibration	Lower calf gain; stronger shafts
Encoder	Magnet rubbing / over-current	Single-encoder arrangement
Hip ODrive	Protection events	Velocity-limit + firmware update
Alignment	Rear toe-in	Joint zero-offset calibration
Initialization	Large policy transient	Match learned standing pose

B. Learning and Deployment Architecture

The initial controller was trained with PPO using task rewards for velocity tracking, body orientation, contact pattern, action smoothness, effort, foot clearance, and slip. For the gait-guided controller, the same task objectives were retained and reference rewards were added for joint position, joint velocity, and foot position.

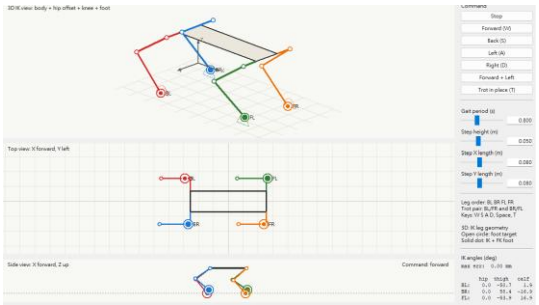


Fig. 2. Interactive gait-generator interface used to visualize and tune the robot-specific reference gait.

Before RL training, the reference was checked with a standalone Python tool that applied the robot-specific kinematics and allowed gait period, step height, and step length to be tuned. The resulting joint trajectories were then executed directly on the physical robot without an RL policy. The robot performed the periodic gait without the catastrophic foot-dragging and actuator-protection failure seen with Pure RL. This was treated only as a qualitative feasibility check, indicating that the selected reference lay within or near a hardware-executable motion region.

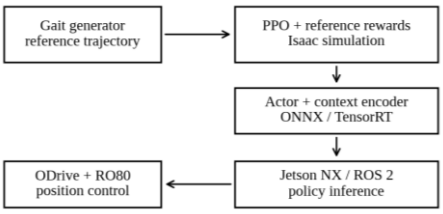


Fig. 3. Gait-guided training and deployment pipeline. The gait generator supplies motion references during PPO training, while the learned networks execute joint commands through the same ROS 2 and ODrive stack used for hardware testing.

Deployment runs on a Jetson NX under ROS 2. The pure-RL model is exported as a policy network, whereas the gait-guided implementation uses an EncoderNetwork together with an ActorNetwork. Both controllers ultimately send joint-position commands to the same low-level ODrive loops. Joint angle, motor command, motor speed, measured and commanded current, IMU data when available, and the remote velocity command are recorded in ROS bags. Using the same low-level hardware stack is important because it lets the observed difference be traced primarily to the learned motion and its interaction with the real mechanism.

III. EXPERIMENTAL METHOD

A. Datasets and Test Conditions

Three data groups are used for quantitative analysis. The Pure-RL set contains one simulation playback and one recorded hardware failure sequence. The gait-guided set contains two trained controllers, each with one simulation playback and three hardware recordings at 0.4 m/s. The direct gait-generator execution in Section II-B is a qualitative feasibility check and is not included in these groups. The retained quantitative records contain synchronized motor and motion data suitable for the same processing pipeline.

TABLE II. TEST CONDITIONS AND DATA COVERAGE

Quantity	Pure-RL	Gait-guided-RL
Learning objective	PPO task rewards	PPO + gait-reference rewards
Reference step height	None	0.10 m / 0.09 m
Simulation gait samples	~20 swing peaks / leg	45–48 / 92–95 peaks / leg
Hardware records analyzed	1 failed deployment record	6 successful records (3 per controller)

Quantity	Pure-RL	Gait-guided-RL
Hardware command	Protection at 0.242 m/s during ramp	0.4 m/s steady command
Hardware outcome	No complete gait cycle in analyzed record	Sustained walking in all 6 analyzed records

B. Encoder-Consistent Foot-Clearance Metric

Simulation provides world-frame foot height, but a synchronized world-frame base pose is not available in every hardware bag. To avoid mixing two definitions, foot motion in both domains was reconstructed from joint position with the same forward-kinematics model. For successful guided walking, the stance baseline of each leg is defined as the 10th percentile of body-frame foot height within the steady window rather than the absolute minimum, because the minimum is more sensitive to encoder noise, small body oscillations, and occasional outlier samples. Here, $z_i^B(t)$ denotes the vertical position of foot i in the body frame at time t, and $Q_{0.10}[z_i^B]$ denotes the 10th percentile of that signal over the analyzed steady segment:

$$c_i(t) = z_i^B(t) - Q_{0.10}[z_i^B] \quad (1)$$

where $c_i(t)$ is the stance-referenced clearance of foot i; values near 0 cm correspond to the stance baseline and positive values indicate swing elevation. For the guided trials, swing peaks were detected from the reconstructed clearance signal and cycles were resampled to 0–100% phase. Model-level hardware values are the mean of three independent deployments. The main amplitude measures are peak retention and the mean absolute leg-wise peak gap:

$$R_h = 100 \cdot \text{mean}_i(h_i^{\text{real}}) / \text{mean}_i(h_i^{\text{sim}}),$$
$$E_h = \text{mean}_i\, |h_i^{\text{sim}} - h_i^{\text{real}}|. \quad (2)$$

where h_i^{sim} and h_i^{real} are the mean swing-peak clearances of leg i in simulation and hardware, respectively, and mean_i denotes averaging across the four legs. Phase RMSE and Pearson correlation are computed from the averaged normalized cycles. Joint tracking RMSE compares motor_cmd with encoder feedback after time interpolation, and current RMS/peak current are taken from motor_current_measured. The pure-RL deployment failed before a full gait cycle; it is therefore analyzed with the maximum pre-protection foot displacement, time to the current-limit event, and front-thigh command/feedback error rather than with steady-cycle metrics. The FSR channel in the pure-RL bag remained zero, so ground-reaction and friction forces during toe drag are not quantified directly.

IV. RESULTS AND DISCUSSION

A. Pure-RL Sim-to-Real Failure

The pure-RL simulation did not reveal the severity of the hardware problem. Using the same encoder-consistent forward kinematics, the simulated mean swing peak was 7.54 cm; the FL and FR peaks were 7.31 and 6.65 cm. The policy therefore appeared to provide enough geometric clearance to attempt deployment. The recorded hardware test behaved very differently. Before the current-limit event, the largest FL and FR foot displacements were only 2.57 and 3.53 cm, and no complete walking cycle was formed. These partial-cycle values are not reported as a formal retention ratio, but their scale makes the low-clearance failure clear.

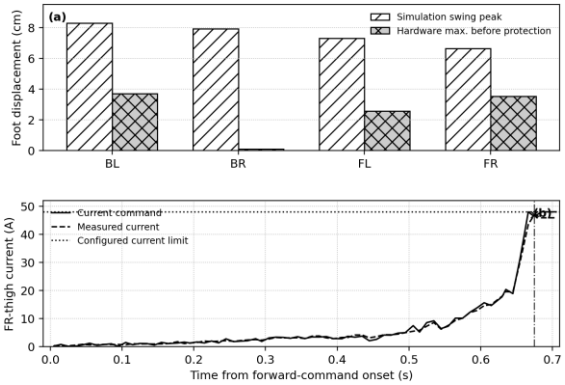


Fig. 4. Pure-RL Sim-to-Real failure. (a) Simulated swing peaks and the maximum hardware foot displacement observed before front-leg protection; no complete hardware gait cycle was obtained. (b) FR-thigh current during the command ramp, approaching the configured 48 A limit about 0.675 s after command onset.

TABLE III. PURE-RL DEPLOYMENT FAILURE: SIMULATION AND HARDWARE EVIDENCE

Metric	Simulation	Hardware attempt
Foot motion	Mean swing peak: 7.54 cm	No complete gait cycle
FL / FR foot motion	7.31 / 6.65 cm swing peak	2.57 / 3.53 cm max. before protection
FR-thigh load	25.1 N·m torque RMS	47.99 / 46.98 A command/ measured
FR-thigh tracking	12.9 deg target RMSE	11.8 deg RMSE; 33.8 deg max. error
Torque / current limit	28 N·m; 70.1% of steady samples at limit	Current event at 0.242 m/s command
Time / outcome	Sustained simulated gait	Protection after 0.675 s; no forward walk

The current trace explains why the failure escalated quickly. The FR-thigh current command reached 47.99 A and the measured current reached 46.98 A, essentially the configured 48 A limit, only 0.675 s after the forward command began. Over the same pre-protection interval the FR-thigh tracking RMSE was 11.8 deg and the instantaneous error reached 33.8 deg. After the event, the front-leg current channels stopped changing while the rear-leg channels continued to update in the bag, consistent with the protection/disarm observed during the experiment. The robot then pitched forward and the front legs lost support.

The likely load path is also consistent with the observed toe drag. A foot that does not enter a clean swing phase continues to carry normal and frictional reactions from the floor; those contact reactions add to the joint effort required at the thigh. The force sensor in this particular bag did not provide usable nonzero data, so this mechanism is not presented as a direct force measurement. However, the simulation itself already showed little actuator margin: after the startup transient, the FR thigh was at the 28 N·m applied-torque limit in 70.1% of the recorded samples. Small unmodeled contact, friction, or transmission losses could therefore push the real actuator into current protection.

B. Training and Actuator Diagnostics

The three training logs provide a useful caution about judging transfer from a single simulation score. In the final 100 logged iterations, the pure-RL run had a lower mean velocity-error metric (0.236) than the two gait-guided runs (0.513 and 0.530). Nevertheless, its mean episode length was only about 595 steps, while both guided runs were essentially at the 1000-step horizon. The logged base-contact termination signal was also much more prominent in the pure-RL run. The reward sets and training histories are not identical, so these values are used as diagnostics rather than as a controlled algorithm benchmark.

The actuator traces provide evidence more directly related to hardware safety. After startup, the pure-RL FR thigh remained at the 28 N·m simulated effort limit in 70.1% of the samples, whereas neither guided controller reached its 30 N·m front-thigh limit in the corresponding steady segment; the FL thigh spent 10.6% of pure-RL samples at its limit. Thus, the pure-RL policy combined acceptable simulated foot motion with little torque reserve. The later over-current event is consistent with a simulation-valid policy operating near the physical feasibility boundary, where unmodeled contact and transmission losses can make the same motion unreliable on hardware.

TABLE IV. TRAINING-LOG AND SIMULATION ACTUATOR DIAGNOSTICS

Metric	Pure RL	Guided A	Guided B
Last iteration	3499	6659	7193
Episode length (last 100)	595.5	999.7	999.4
Velocity error (last 100)	0.236	0.513	0.530
FL thigh: torque RMS / at limit	13.5 N·m / 10.6%	7.73 N·m / 0%	5.77 N·m / 0%
FR thigh: torque RMS / at limit	25.1 N·m / 70.1%	4.63 N·m / 0%	10.98 N·m / 0%

C. Control Interpretation of the Training-Strategy Change

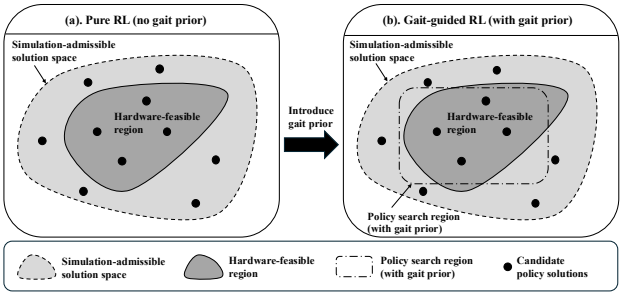


Fig. 5. Conceptual illustration of the feasibility mismatch. Gait guidance narrows the effective policy-search region and increases its overlap with hardware-feasible motions. Candidate solutions are schematic; the figure is not to scale.

Figure 5 conceptually illustrates our interpretation of the Pure-RL failure as a mismatch between a broad simulation-admissible solution space and a smaller hardware-feasible region. Assembly tolerances, drivetrain losses, backlash, contact uncertainty, actuator limits, and sensing errors can make some simulation-valid motions fragile on the physical robot. Because it was impractical to make the full range of simulation-admissible solutions reliably transferable to hardware, we instead reduced the effective policy-search space. The gait-generator reference was first shown to be directly executable on hardware and was then used as a structured prior during learning. As illustrated in Fig. 5(b), this prior constrains the effective policy-search region toward motions with reasonable swing timing and clearance, thereby increasing its expected overlap with the hardware-feasible region. RL is retained for task optimization and for adaptation beyond the fixed kinematic reference. This interpretation does not claim hardware-optimal torque, acceleration, or efficiency; rather, it explains the design choice used to reduce the likelihood of simulation-specific solutions.

D. Gait-Guided Sim-to-Real Foot Clearance

After the direct gait-generator feasibility check, the same reference was introduced during learning to make swing timing and foot placement explicit while preserving RL adaptation. All six analyzed gait-guided recordings then contained repeated foot lifting and a usable forward-walking interval.

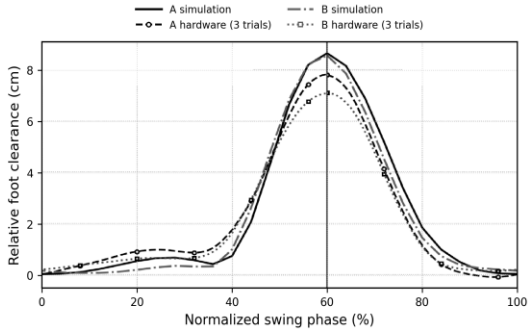


Fig. 6. Phase-normalized swing-clearance trajectories in simulation and hardware. Hardware curves are averaged over three trials per controller.

TABLE V. GAIT-GUIDED SIMULATION-TO-HARDWARE TRANSFER AT 0.4 M/S

Metric	Guided A	Guided B
Sim. mean swing peak	8.63 cm	8.54 cm
Real mean swing peak	7.78 ± 0.06 cm	7.25 ± 0.45 cm
Amplitude retention	90.0%	84.9%
Mean leg-wise peak gap	1.09 cm	1.31 cm
Phase RMSE	0.86 cm	0.73 cm
Phase correlation	0.980	0.990
Real joint RMSE	6.53 ± 1.24 deg	6.86 ± 3.57 deg
Real current RMS	8.41 ± 0.65 A	10.32 ± 1.66 A
Peak |current|	47.93 A	49.42 A

The gait-guided trajectories still lose amplitude after deployment, but the transfer is qualitatively different from the failed Pure-RL case. The 8.63 and 8.54 cm simulated mean swing peaks become 7.78 ± 0.06 and 7.25 ± 0.45 cm on hardware, corresponding to amplitude retentions of 90.0% and 84.9%. At the same time, the phase-normalized correlations remain high (0.980 and 0.990) and the phase RMSE stays below 0.9 cm. This combination indicates that the timing and overall shape of the swing trajectory transfer well, while the main residual error appears as a reduction in vertical amplitude. The repeated trials further show that this is not a single-run effect: all six recordings contain continuous swing cycles and forward walking, with mean peak clearance spanning 6.95–7.85 cm.

E. Residual Joint, Body, and Load Gaps

The remaining clearance loss is not explained by the policy output alone. Hardware joint-position RMSE is 6.53–6.86 deg overall, with the calf at about 9.65 deg in both guided conditions. Because the calf contains the additional 2:1 transmission stage, its higher error is consistent with the effects of backlash, friction, compliance, and gain mismatch introduced below the policy level. Current RMS remains 8.41 ± 0.65 A and 10.32 ± 1.66 A across the two guided datasets, while brief peak currents can still approach the protection range. The important difference from the Pure-RL failure is therefore not the complete absence of high-current samples, but that the guided gait does not remain trapped in a high-load contact condition: the feet establish repeated swing phases and the robot continues to advance.

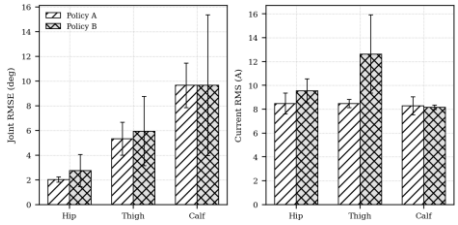


Fig. 7. Hardware joint-position tracking RMSE and motor-current RMS by joint family for the two gait-guided controllers. Values are mean ± standard deviation over three deployments.

Body orientation shows an even larger domain shift than foot trajectory. Roll standard deviation increases from 0.83 deg in simulation to 3.09 ± 0.22 deg on hardware for Guided A and from 0.53 deg to 5.31 ± 0.77 deg for Guided B, rising by approximately 3.7-fold and 10.0-fold for the two guided controllers. Pitch variation similarly increases from 0.57 deg to 2.77 ± 0.07 deg and from 0.39 deg to 3.39 ± 1.38 deg, rising by about 4.9-fold and 8.7-fold. Figures 6 and 8 make this contrast explicit: the swing waveform retains high phase correlation, but body-level dynamics are substantially less quiet on hardware. This indicates that gait guidance preserves the geometry and timing required for walking without eliminating residual drivetrain, disturbance-response, and state-estimation differences. Quaternion data were available for two of the three trials per controller, so the orientation statistics are reported separately from the six-trial foot, joint, and current results.

TABLE VI. SIMULATION-TO-HARDWARE BODY-ORIENTATION DISPERSION

Metric	Guided A	Guided B
Roll SD (sim / real)	0.83 / 3.09 ± 0.22 deg	0.53 / 5.31 ± 0.77 deg
Pitch SD (sim / real)	0.57 / 2.77 ± 0.07 deg	0.39 / 3.39 ± 1.38 deg

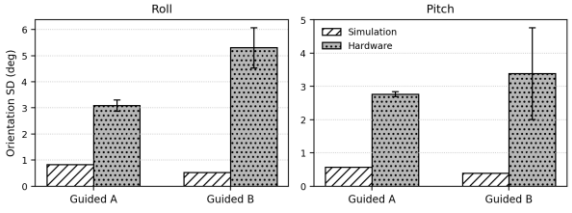


Fig. 8. Simulation-to-hardware body-orientation dispersion. Bars show roll and pitch standard deviation; hardware error bars indicate variation across the two trials with valid quaternion data.

Repeated deployments show why successful transfer should not be summarized by a single run. Across all six analyzed records, mean swing peak ranges from 6.95 to 7.85 cm, all-joint RMSE from 3.40 to 10.52 deg, and current RMS from 7.90 to 11.94 A. Figure 9 shows the trial-level spread. Guided A is particularly repeatable in clearance, with three means clustered between 7.73 and 7.85 cm, whereas Guided B spans 6.95–7.76 cm and also shows larger variation in joint RMSE and current RMS. Despite this run-to-run variation, every guided record retains periodic swing and sustained forward motion.

The repeated trials also separate systematic transfer bias from experimental variability. Hardware peak clearance is lower than the corresponding simulated value for both guided controllers, so the direction of the amplitude error is shared across policies rather than being produced by one atypical run. The evidence therefore supports two conclusions at once: under the tested 0.4 m/s level-ground condition, gait guidance avoided the observed catastrophic Pure-RL failure mode, while a persistent platform-level amplitude gap remained after transfer.

TABLE VII. SIX ANALYZED GAIT-GUIDED HARDWARE RECORDS

Trial	Guided A: peak / joint RMSE / current RMS	Guided B: peak / joint RMSE / current RMS
1	7.85 cm / 5.81 deg / 9.14 A	7.05 cm / 10.52 deg / 10.39 A
2	7.74 cm / 7.96 deg / 8.20 A	7.76 cm / 6.67 deg / 8.63 A
3	7.73 cm / 5.82 deg / 7.90 A	6.95 cm / 3.40 deg / 11.94 A

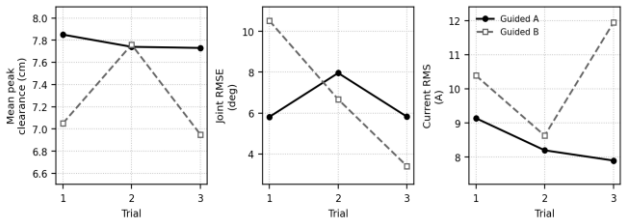


Fig. 9. Trial-level repeatability across the six gait-guided hardware deployments. The three panels show mean swing-peak clearance, all-joint tracking RMSE, and motor-current RMS for the three trials of each guided controller.

F. Interpretation and Next Steps

The results suggest two distinct Sim-to-Real problems. First, apparently acceptable simulation performance can still leave insufficient swing and actuator margin for the physical mechanism. Second, once continuous walking is achieved, a residual-accuracy gap remains in clearance amplitude, body motion, and joint/current behavior.

Further improvement should combine learning changes with hardware identification. Joint-specific friction, backlash, assembly offsets, transmission efficiency, effective gain, current-to-torque behavior, and the additional calf reduction should be identified and represented in simulation. Contact and friction parameters should be validated with reliable force measurements, and the state estimator should provide consistent body-velocity and contact information. On the learning side, the gait reference should remain strong enough to preserve swing geometry while still allowing the policy to adapt timing and amplitude to the real mechanism.

G. Practical Implications for Sim-to-Real Design

A practical implication is that pre-deployment screening should not rely on simulated velocity tracking or foot height alone. For this platform, swing-foot clearance, body-pitch variation, time spent near actuator effort limits, and the available torque/current margin should be checked together. The pure-RL controller would already have been flagged by such a margin-based test because one front thigh remained close to saturation for most of the simulated gait despite apparently acceptable foot motion.

The remaining 10–15% clearance-amplitude loss and the larger hardware orientation dispersion point to the next layer of model improvement: hardware-informed randomization of drivetrain and contact parameters, together with more consistent body-state and contact estimation. These changes target the uncertainties that actually appear in deployment rather than compensating for them indirectly through larger reward weights [4], [10], [11].

Structured gait guidance should therefore be retained while actuator realism and feedback estimation are further improved. Direct gait-generator execution is useful as a feasibility check, but a fixed kinematic gait alone does not provide the adaptive capability sought from learned control. The reference is therefore used as a hardware-executable motion prior: it narrows early policy search toward a realizable neighborhood while RL remains responsible for task optimization and later adaptation. This division of roles separates feasibility assurance from performance optimization, which is the central design principle explored in this study.

The present experiments are limited to 0.4 m/s straight, level-ground forward walking; variable-speed commands, turning, and uneven-terrain locomotion remain future work. The pure-RL hardware result is a documented failure trial rather than a repeated steady-state baseline, because the controller never reached steady walking. For this reason, the paper does not assign a formal pure-RL versus guided percentage improvement. The evidence is instead twofold: the pure-RL model failed before a complete cycle and triggered front-leg protection, while the gait-guided models completed six walking trials and retained 84.9–90.0% of their simulated mean swing amplitude.

V. CONCLUSION

This work began with a practical Sim-to-Real failure on a custom quadruped robot. A Pure-RL controller produced a repeatable simulated gait but could not establish a complete hardware walking cycle because of insufficient clearance, rapidly increasing front-thigh load, and over-current protection. Based on this failure, we constrained learning around a robot-specific gait-generator trajectory that had first been verified by direct hardware execution, using it as a structured PPO motion prior rather than as a replacement for RL. Under the evaluated 0.4 m/s straight, level-ground forward-walking condition, the guided controllers sustained all six analyzed hardware trials, retaining 84.9–90.0% of simulated mean swing amplitude with 0.980–0.990 phase correlation. Residual clearance, body-orientation, joint-tracking, and current gaps remain. These results support hardware-executable gait guidance as a practical way to reduce transfer failure risk on this platform; broader speed, turning, and terrain claims require future evaluation.

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